

North South University
Department of Electrical and Computer Engineering

CSE299

Final Report

Section: 09

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Introduction:

The rapid advancement of technology has led to a significant increase in the use of online portals to manage academic and administrative tasks for university students. While online portals offer several advantages, such as convenience, flexibility, and accessibility, students often face challenges in navigating these portals, leading to frustration and decreased productivity. Moreover, universities also face challenges in managing their online platforms, leading to inefficiencies and delays in providing timely feedback and support to students. To address these challenges, we propose to develop an online portal management system that will offer a user-friendly interface for students to access their academic and administrative tasks easily. This system will also provide a comprehensive solution for universities to efficiently manage their online platforms, including maintaining and updating course information, tracking student progress, and providing timely feedback and support. Our proposed online portal management system will leverage the latest technology and user experience design to create a seamless and efficient experience for students and universities. The system will offer a range of features, including course registration, tuition fee payments, course materials, assignment submissions, communication with professors and classmates, and academic progress tracking. This proposal aims to outline the need for an online portal management system for university students and the potential benefits of such a system. With the proposed system, universities can better manage their online platforms, provide better support and feedback to students, and improve academic outcomes. On the other hand, students can enjoy a more user-friendly and efficient experience when managing their academic and administrative tasks.

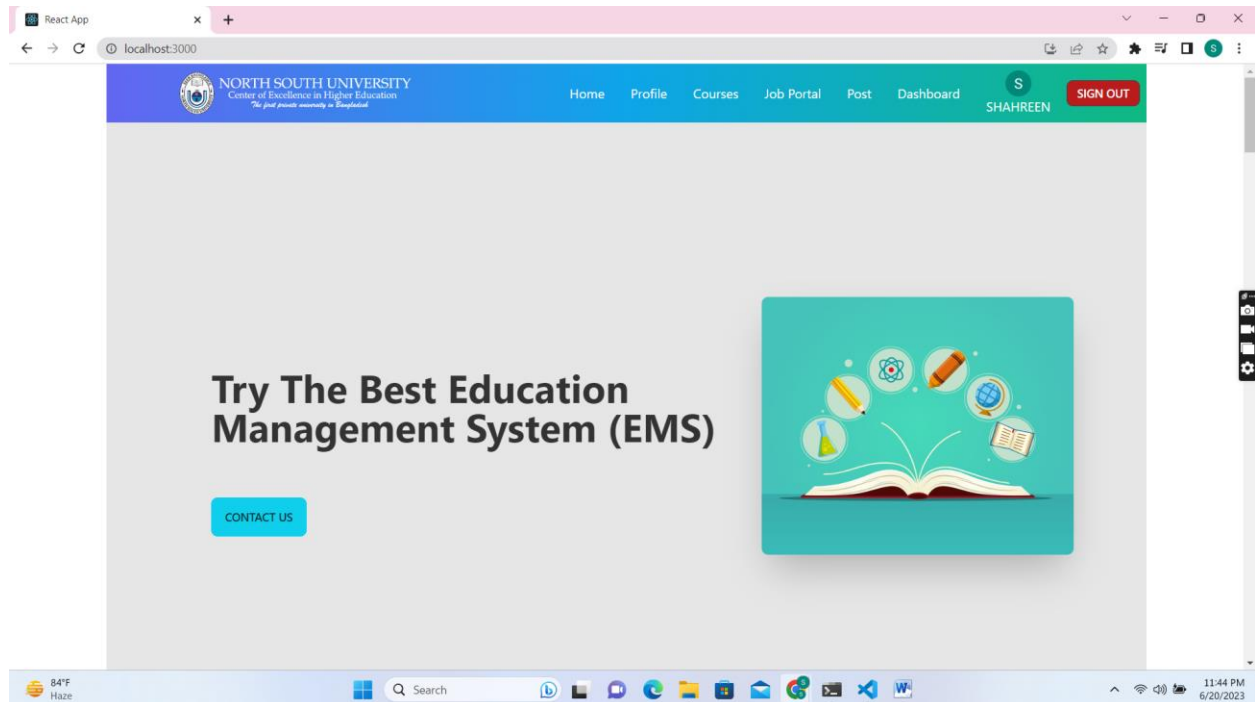
Project Statement:

University students have a wide range of academic and administrative tasks that they need to complete through an online portal. These duties include enrolling in classes, paying tuition, obtaining course materials, turning in assignments, corresponding with instructors and peers, and keeping tabs on academic performance. Unfortunately, a lot of students find it difficult to use these online resources, which causes frustration, uncertainty, and a drop in productivity. Students who are new to the university or who have no prior experience using online platforms may find this to be particularly difficult. Moreover, universities also face challenges in managing their online portals, including maintaining and updating course information, tracking student progress, and providing timely feedback and support to students. Therefore, there is a need for a user-friendly and comprehensive online portal management solution that can help university students efficiently navigate their academic and administrative tasks and help universities better manage their online platforms to improve the student experience and academic outcomes.

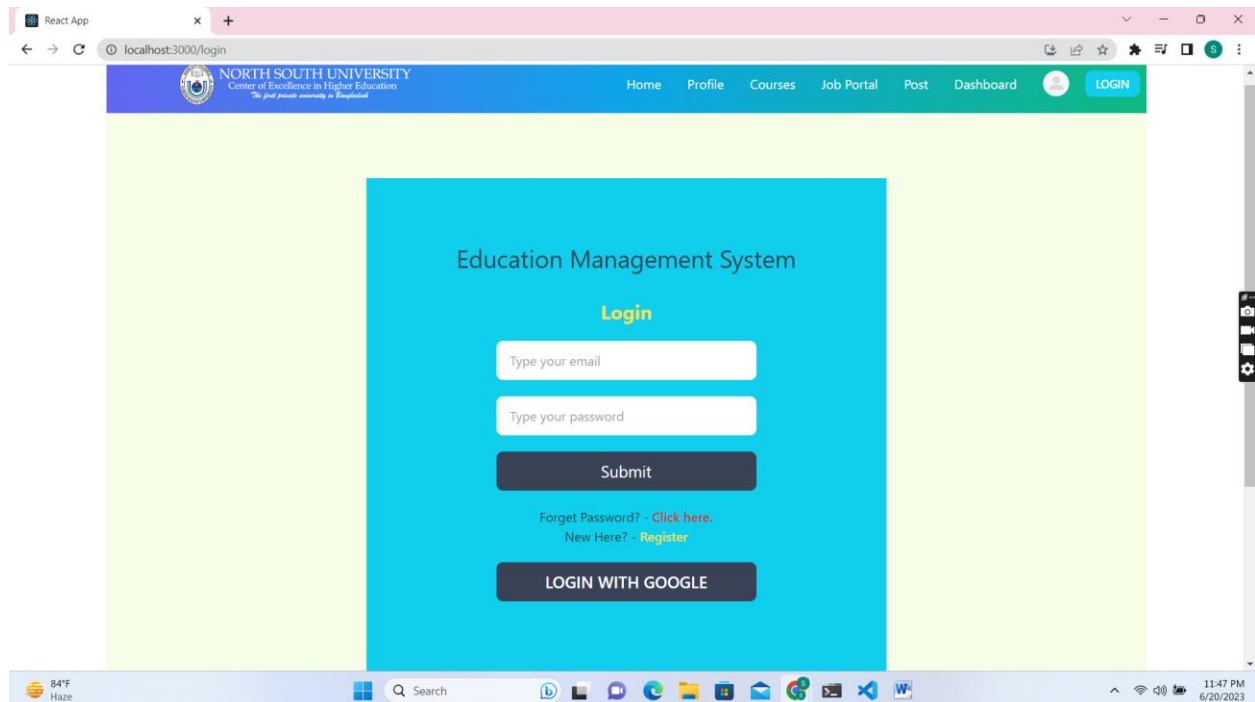
Project Description:

The Online Portal Management for University Students project is aimed at developing a comprehensive web-based platform that will serve as an online resource center for university students. The platform will be designed to offer a wide range of functionalities that will enable students to access essential academic and non-academic information, interact with their peers, and stay up-to-date with university events and news. The portal will have a user-friendly interface that is easily accessible from any device, making it convenient for students to access the platform from anywhere and at any time. The project will require a team of skilled developers and designers to develop the platform's features, and it will be critical to ensure that it is secure, reliable, and scalable to accommodate the growing number of users. It also makes enrolling easier for students by showing them details of any course and online payment options. Overall, the Online Portal Management for University Students project will help to improve the student experience and provide a centralized platform for students to access essential information and resources. Here are the features of the project:

Home Page:

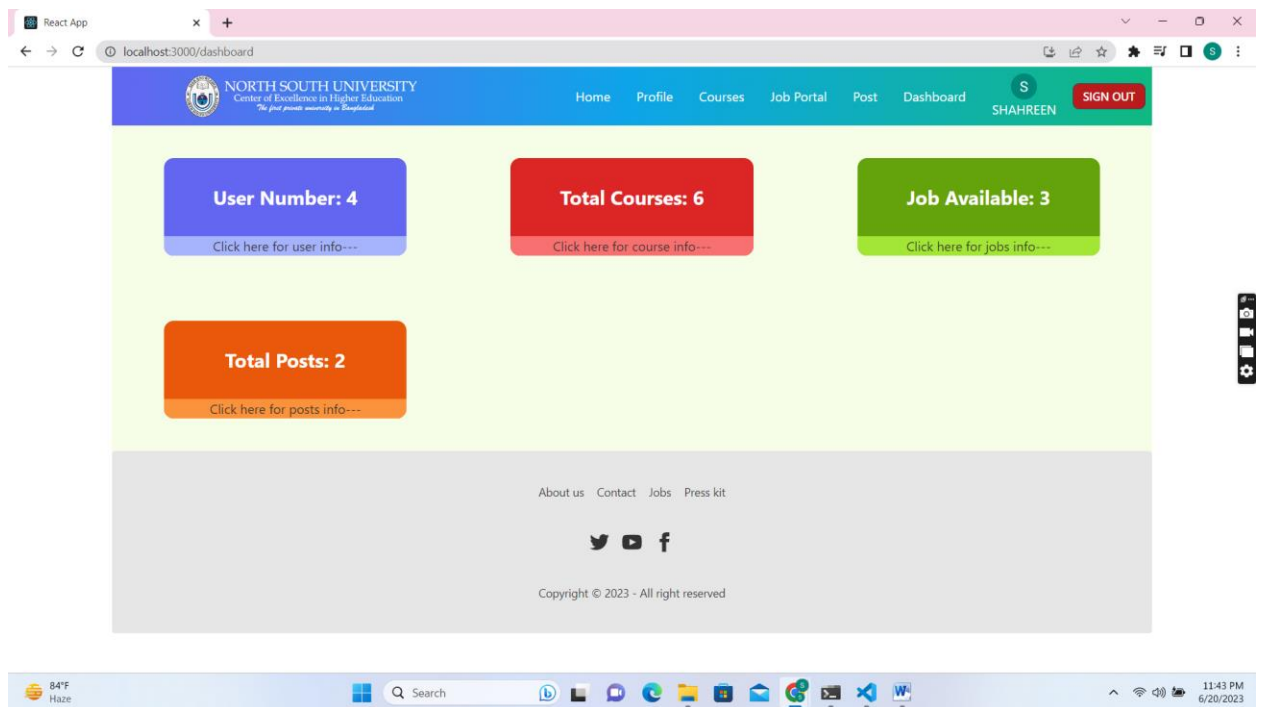


Login:



Teacher Login:

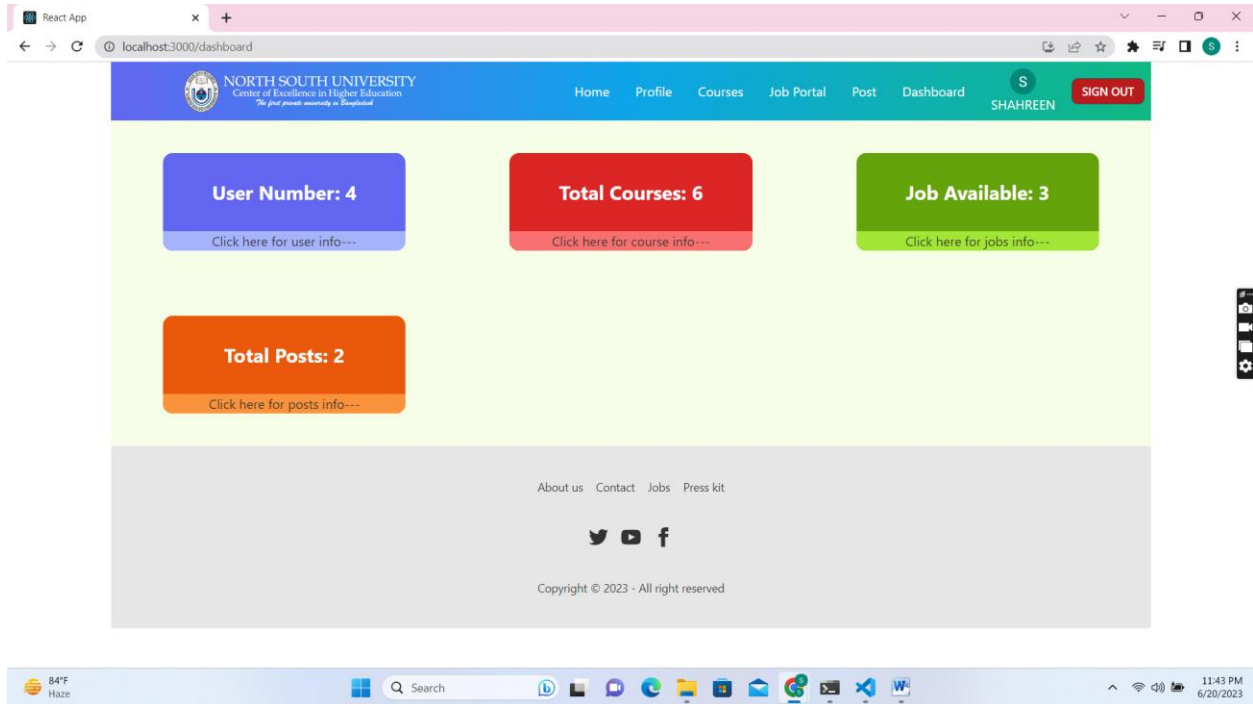
- Dashboard



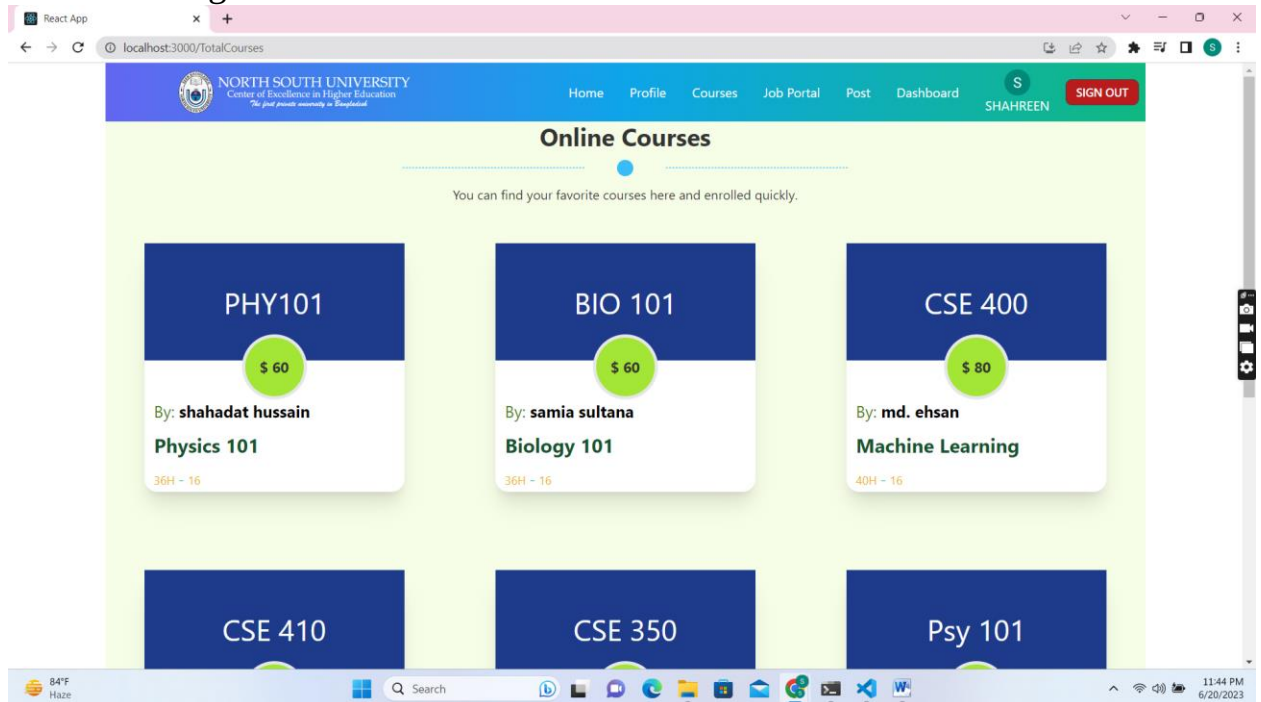
- Course Management
 - Add and Remove Students
 - Course Scheduling
 - Communication
 - Collaboration
 - Progress Tracking

Student Login:

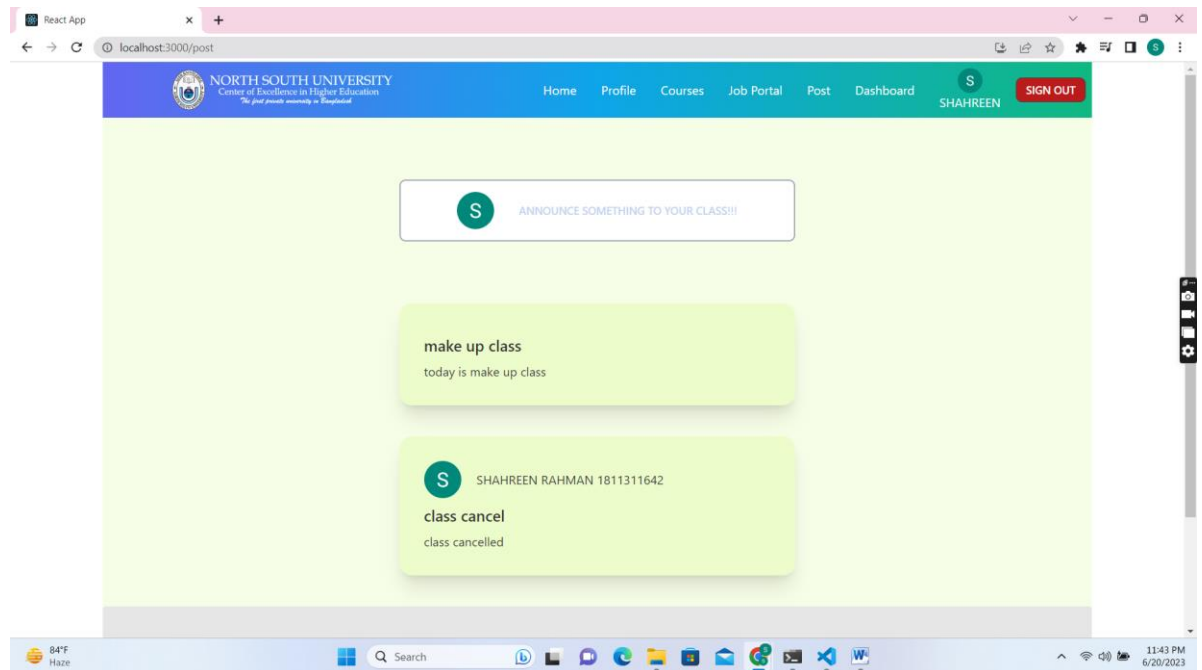
- Dashboard



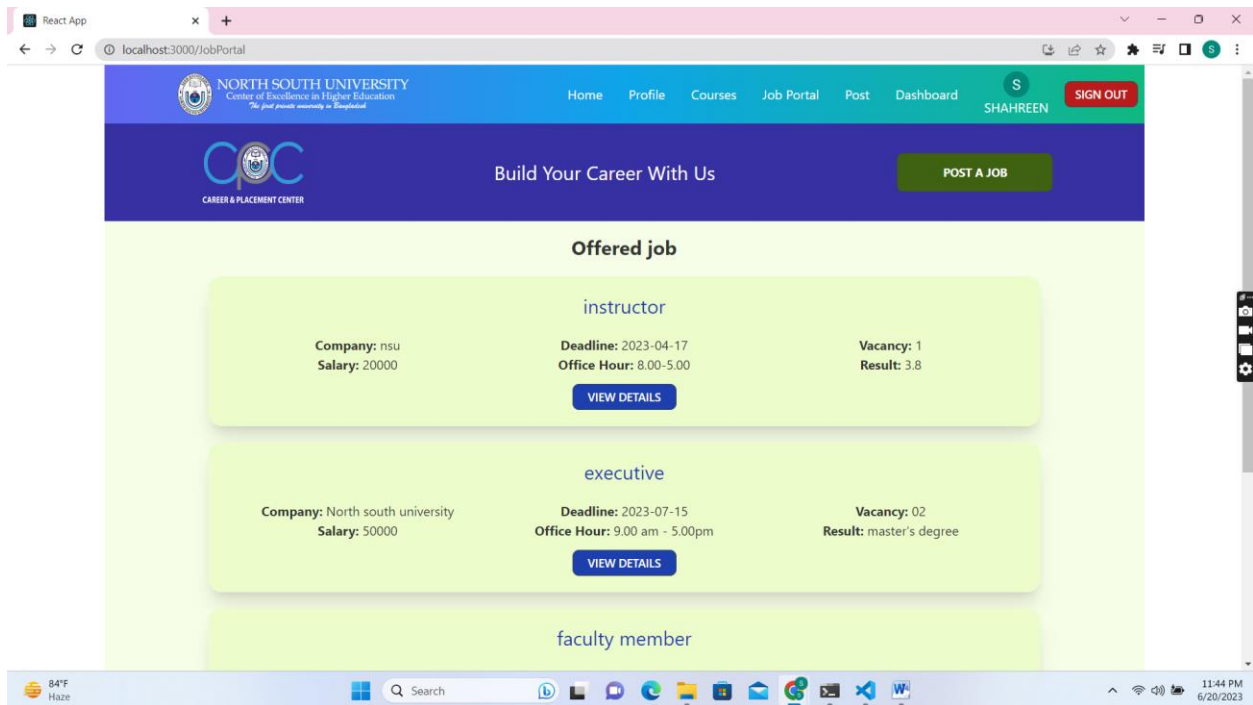
- Course Management



- Events Calendar
- Personalization
- Attendance Tracking
- Online Courses
- Online enrollment payment
- Academic Advising
- Posts

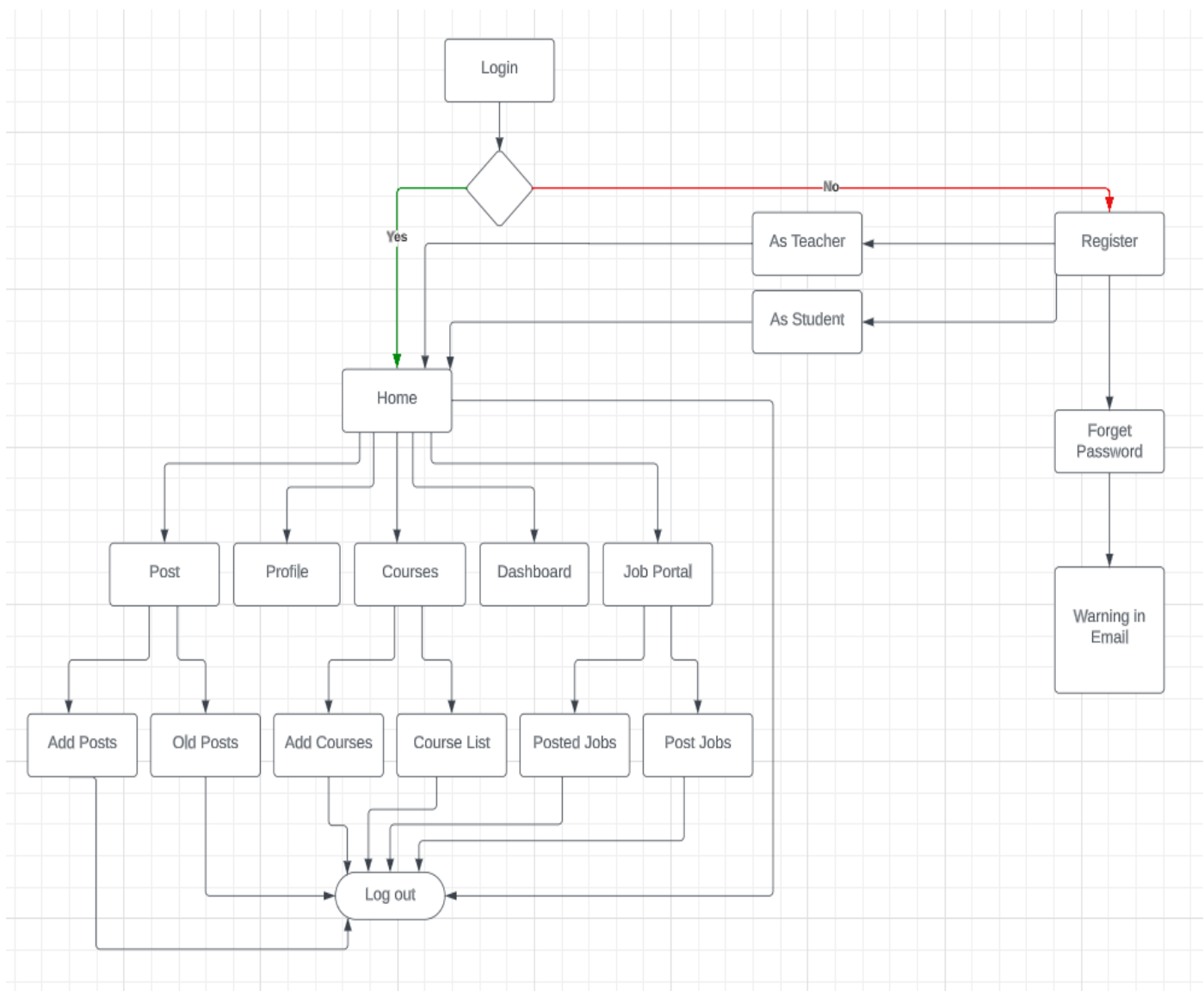


- Job and Internship Opportunities



- Personal Development
- Accessibility

Project Flowchart:



Technologies used:

- Node js:
Node.js is a back-end JavaScript runtime environment. Mainly used for working with the backend of our application. We used it to make the backend framework of our project.
- React js:
React is a JavaScript library for building user interfaces based on components. We used it to build the front-end of our website.
- MongoDB:
MongoDB is a database program. It is a NoSQL database program, MongoDB uses JSON-like documents with optional schemas. We used it for our database.
- Firebase:
Firebase is a set of backend cloud computing services and application development platforms. It hosts databases, services, authentication, and integration for a variety of applications. We used it to authenticate our website. Each email registered in the website will be stored in Firebase.

Conclusion:

We made this website to make communication between teachers and students easier. This makes students to communicate with their peers, which in turn increases their performance in academics. Teachers can communicate and get feedback from their students. They can also watch over their students' progress in enrolled courses. It also makes enrolling easier for students by showing them details of any course and online payment options. This makes it a useful and necessary website for any educational institute to have.

For easier access to our website we uploaded it in GitHub.

GitHub link for server-end:

<https://github.com/shahreen-simin/education-website-server>

GitHub link for client-end:

<https://github.com/shahreen-simin/education-website>